PD^{2}Net: A Prototype-Guided Generative Copy-Move Forgery Image Detection and Distinguishment Framework

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***Abstract*—Emerging developments in image editing have raised substantial hurdles regarding the reliability of multimedia data, which promotes the exploration of forgery image detection. Copy-move forgery, a prevalent tampering technique, is defined as duplicating one area of an image (source region) to another area of the same image (tampered region). Consequently, the source and tampered regions exhibit identical data distributions, posing a formidable obstacle to simultaneously detecting and distinguishing them. Existing discriminative methods struggle to distinguish source and tampered regions due to the absence of explicit prior constraints, especially under identical data distributions. To address this limitation, we propose a novel generative framework, PD^{2}Net, which formulates the task as a role-constrained representation construction process guided by learnable prototypes, enabling end-to-end detection and differentiation of source and tampered regions. Specifically, two distinct prototypes are introduced as priors for the source and tampered regions, and are iteratively optimized through prototype-feature interaction to progressively refine role-constrained representations. This mechanism alleviates the difficulty caused by identical feature distributions and enables more stable distinguishment. In addition, we propose Suspect Region Selection (SS) and Suspect Region Re-identification (SR) modules to obtain the object-level refined target guided by the prototype by eliminating false positive pixels. Extensive experiments on USC-ISI, CASIA v2.0, and CoMoFoD datasets confirm the effectiveness and robustness of our proposed framework in the field of copy-move forgery image detection and distinguishment.**

***Index Terms*—Copy-move forgery image detection, deep matching, edge supervision, refinement.**

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I. Introduction

DIGITAL images have become a dominant medium for representing and disseminating information. However, the rapid advancement of image editing technologies has enabled malicious manipulations, such as fabricating evidence or spreading misinformation, posing significant threats to public trust and digital security. Consequently, reliable methods for image forgery detection are imperative for multimedia forensics [1].

Copy-Move, as a common tampering method, is defined as the process of copying one part in an image (the source region) and pasting it to another location in the same image (the tampered region) [2], [3]. The existing detection tasks for copy-move forgery images, which are illustrated in Fig. 1, can be categorized into three types: 1) image-level detection, which determines whether the image has been tampered [4], [5]; 2) pixel-level detection, a binary classification task that detects a pair of similar areas and backgrounds [6], [7], [8]; 3) pixel-level detection and distinguishment, a three-classification task that distinguishes the source region and tampered region based on task (2) [9], [10].

For the pixel-level task, the existing copy-move focuses on task (2), which uses self-correlation to detect highly similar regions in images. However, detecting only a pair of similar regions without distinguishing them fails to address practical needs [11]. Distinguishing the source and the tampered region in the copy-move forgery image can reveal the motive and means of tampering, which helps carry out more in-depth criminal investigation and forensics [12]. However, the distinguishment task faces a challenge in the copy-move forgery image, where the tampered region replicates from the source region, and both have the same data distribution. Therefore, the direct application of binary classification tasks for distinguishment fails to achieve satisfactory outcomes.

Existing methods for detecting and distinguishing between source and tampered regions fall into four paradigms [11], [12], as shown in Fig. 2. The first paradigm adds tampered region detection branches and fuses them with similar region detection branches to realize distinguishment. This method only superimposes the features of the tampered region onto the features from similar areas. Still, it does not provide a definite distinguishment between the source and the tampered region. Therefore, it can only optimize the results but cannot essentially achieve the

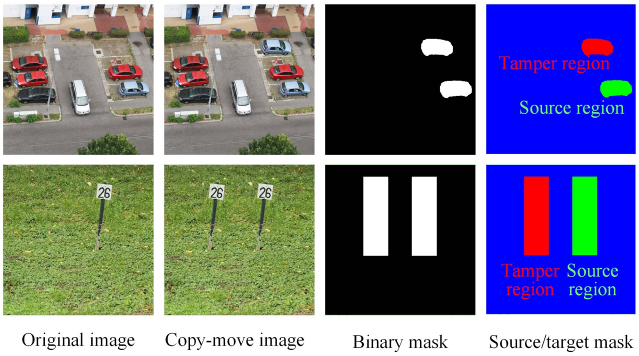

Fig. 1. Some examples of forgery image detection. The first column shows the original image, the second column shows the tampered image, and the third column shows the detection mask, in which green represents the source region, red represents the tampered region, and blue represents the pristine region (background).

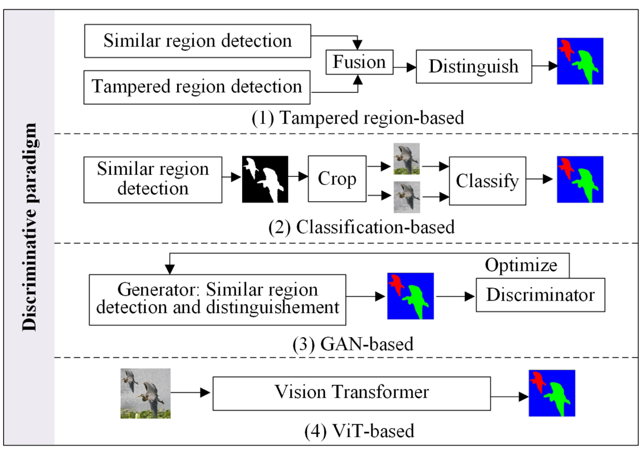

Fig. 2. The existing discriminative paradigms for detecting and distinguishing between source and tampered region in the copy-move forgery images.

distinguishment. The second classification-based paradigm adopts a two-stage process. The first stage locates similar regions, and then these regions are cropped out and input into the classifier for classification in the second stage. This approach focuses on detecting a pair of similar regions while lacking the modeling of the distinguishment between the source/tampered regions. Moreover, it does not have the end-to-end modeling capability that belongs to a posteriori discrimination operation and essentially relies on the detection accuracy of the first stage. The classification result may also become unreliable if the regional detection is inaccurate. The third GAN-based paradigm usually introduces a discriminator based on the traditional similar region detection module and improves the model performance through the adversarial training mechanism. Although this structure can optimize the detection effect to a certain extent, its performance gain mainly stems from the introduction of the discriminator rather than the explicit distinguishment of the source/target area. In addition, as the fourth paradigm, the transformer-based image processing method has been employed in copy-move forgery distinguishment tasks due to its superior capability in modeling long-range dependencies [13]. While the above methods achieve certain success, they predominantly rely on feature-level implicit discriminative paradigm and fail to obtain edge-structurally complete and object-level distinguishment results. More fundamentally, these methods lack an explicit mechanism to model the roles of source and tampered regions, and instead depend on indirect feature separability, which leads to ambiguity under identical data distributions.

In contrast, PD^{2}Net adopts object-level explicit generative paradigm, where each region is assigned to a predefined role under explicit role constraints via prototype-feature interaction. By explicitly modeling the role assignment process, PD^{2}Net addresses the challenge of distinguishing source and tampered regions with identical data distributions from a fundamentally different perspective. Furthermore, the proposed method refines detection targets and achieves structurally complete object-level distinguishment.

To address these limitations, we propose PD^{2}Net, a prototype-guided, generative framework for end-to-end detection and distinguishment of copy-move forgery. Instead of relying solely on feature-level implicit discriminative cues, we introduce two learnable prototypes that serve as explicit object-level priors for the source and tampered regions. These prototypes are iteratively optimized together with coarse similarity features, enabling the network to progressively refine its understanding of both regions under identical distributions.

It is important to clarify that PD^{2}Net is not a generative model in the classical sense such as likelihood-based or data synthesis models. Instead, "generation" in our framework refers to the iterative construction and refinement of role-aware, prototype-based representations via prototype feature interaction.

In addition, we design two dedicated modules: the Suspect Region Selection (SS) module, which identifies inconsistencies between prototype-guided features, and the Suspect Region Re-identification (SR) module, which maps inconsistency regions to shallower features for fine-grained re-detection. These components collaboratively reduce false positives and enhance structural completeness, yielding more accurate and interpretable detection outcomes. Through extensive experiments on USC-ISI, CASIA v2.0, and CoMoFoD datasets, we demonstrate that PD^{2}Net outperforms state-of-the-art methods in both detection accuracy and robustness, setting a new benchmark for prototype-based forgery detection frameworks. Our contributions are outlined as follows:

- We propose a prototype-guided framework that performs object-level explicit assignment for copy-move forgery detection and distinguishment, where two learnable prototypes model the source and tampered regions and enforce role constraints via prototype-feature interaction under identical data distributions.
- We propose prototype-guided SS and SR modules to re-detect ambiguous regions and supplement high-confidence areas, thereby refining object-level detection with structurally complete outputs.
- We conduct extensive experiments across multiple benchmark methods, demonstrating that PD^{2}Net achieves state-of-the-art performance in distinguishing source and tampered regions.

II. Related Work

Copy-move forgery image detection can be divided into three methods, namely traditional block-based algorithms [14], [15], [16] and key-point-based detection algorithms [17], [18], [19], as well as currently widely used deep learning-based detection algorithms [20], [21], [22].

The block-based detection algorithm divides the image into multiple blocks, which can overlap or not. The block-based detection algorithm uses discrete wavelet transform (DWT) [23], Fourier transforms (FFT) [24], and other methods to extract the features of the divided block and calculate the similarity between block features to locate tampered regions through locality sensitive hashing (LSH) [25], Kd-tree Block matching algorithms [26], PCA [27], Random Sample Consensus (RANSAC) [28], etc. The detection method based on image blocks can obtain effective detection results, however, it requires abundant computing resources and is not effective for the detection of distorted regions in the image and tampered regions that have undergone geometric changes [29], [30], [31], [32], [33], [34], [35], [36], [37], [38].

The key-based detection algorithm is to detect local feature points, that is, points in high-frequency areas of the image, such as corner points, and describe their characteristics by SURF [39], SIFT [40], etc. Second, the matching of feature points is realized by approaches such as g2NN, and the tampered region is determined according to the matched similar feature point pairs. The feature matching method is based on selecting key points for matching, which can detect geometrically transformed regions with high robustness and reduce the time complexity of calculation. But it has the disadvantage of being unable to detect smooth forged regions [41], [42], [43], [44], [45].

With the development of neural networks, the copy-move detection approach based on deep learning prevails. Compared with the traditional method, the method based on deep learning does not need to extract features manually. It can extract advanced semantic features that are difficult to obtain by conventional methods. The method based on deep learning utilizes a large amount of data for model training, which has stronger robustness and generalization [46].

Rao et al. [47] extracted high-frequency noise through an SRM filter to suppress untampered image content and carried out deep feature extraction through a convolutional neural network to mine the traces left by the tampered region. This is actually a further convolutional feature extraction based on manual feature extraction. Liu et al. [48] abandoned manual features and extracted features only through deep convolutional networks. Unlike determining similarity thresholds based on manual settings or heuristic methods, Wu et al. [49] proposed a soft decision strategy. Specifically, Wu et al. performed correlation calculations on the features extracted by the CNN to obtain the correlation between each pixel and all other pixels. Then, the correlation matrix was sorted row by row and restored into feature representations, where similar regions in the features were highlighted. With the widespread adoption of dense networks and multi-scale techniques, this approach shows significant potential in enhancing tampering detection performance. Zhong et al. [50] proposed a copy-move forgery detection method, utilizing three dense connection feature extraction blocks for multi-scale feature extraction, ensuring the preservation of shallow layer information and fusion of correlation features from different scales to generate detection results. Although this method uses the structure of a feature pyramid, it still has some limitations in acquiring feature dependencies at a long distance. CAMU-Net [51] utilizes a four-stage architecture for copy-move forgery detection, including hierarchical feature extraction, multi-scale forgery area prediction, feature enhancement to suppress irrelevant information, and multi-scale information fusion. This design ensures robust detection by capturing multi-scale features, optimizing matching accuracy, and leveraging both global and local information. Zhu et al. proposed ARNet [52], a residual refinement network utilizing adaptive attention to model global correlation from channel and pixel dimensions, subsequently refining detection results through residual structures. To ensure the integrity of the detection of the target region and eliminate misjudgment, Liu et al. [8] proposed a two-stage model in which the first stage obtains the candidate target and then locates the bounding box with similar objects and carries out the key point matching of the candidate target to correct error detection. To obtain structurally complete detection objects, Wang et al. [53] proposed DRNet, which detects coarse similar regions using a high-order correlation modeling mechanism to ensure detection stability. Additionally, DRNet employs layer-by-layer decoupling to utilize shallow detail features, extracting suppressed similar regions from shallow layers and integrating them into the coarse similar regions to achieve edge-complete detection targets. As transformer-based methods have gained prominence in computer vision, researchers have begun exploring their potential for copy-move forgery detection tasks. Liang et al. [54] proposed LBRT, which uses a ViT-based structure encoding on the global blocks segmented from the image and the local blocks re-segmented. The self-attention features from the two branches are precisely fused, then upsampled and decoded. Therefore, LBRT considers global semantic information modeling and local detail information refinement.

All the aforementioned methods only locate the source and tampered regions without distinguishing between them, which fails to meet the requirements of real-world image forensics tasks. Differentiating source and tampered region in copy-move forgery images can provide more detailed information and more reliable evidence for forensic analysis, enabling better inference of the forger's intentions. Therefore, distinguishing between source region and the tampered region is essential.

Therefore, Wu et al. proposed Busternet [55], which uses self-correlation calculation to detect similar regions and fuses it with tampered regions extracted by an extra branch to realize distinguishment between the source and the tampered region. However, BusterNet's two parallel branches interact heavily with each other, and a poor performance of either branch will affect the final result. Therefore, to solve this problem, Chen et al. Proposed CMSDNet [56], a series network in which a binary classification task was performed to detect a pair of similar regions. Then, the crop operation was performed on the detected pair of similar regions, and the cropped objects

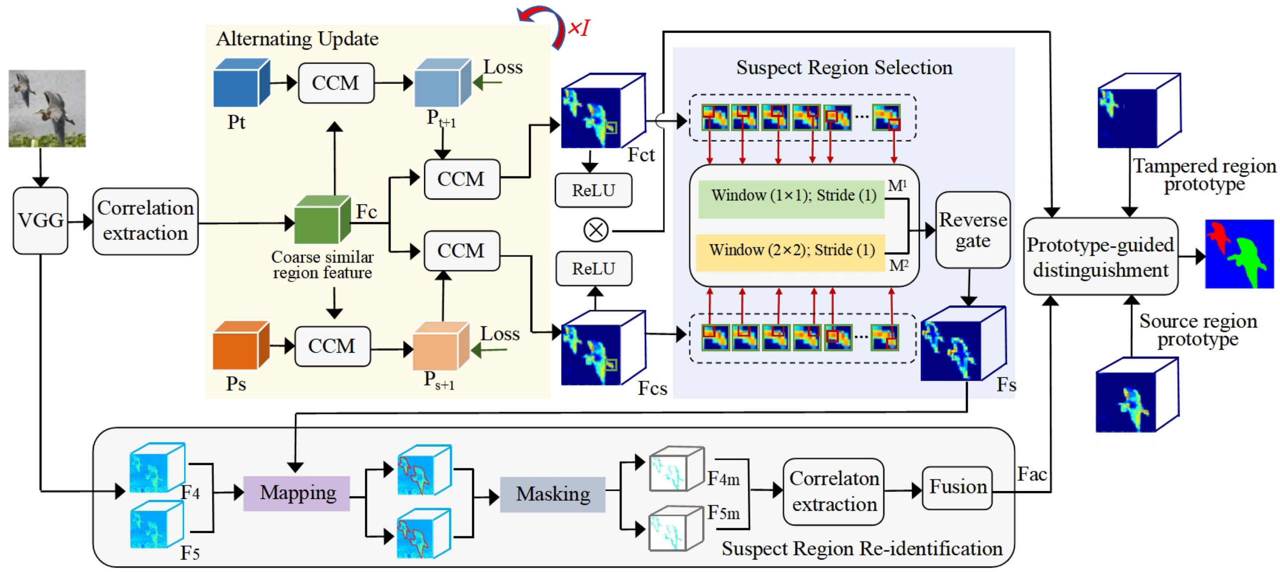

Fig. 3. The overall structure of the PD^{2}Net. PD^{2}Net introduces different prototypes to represent the source and tampered regions and performs cross-correlation modeling with coarse similar regions to obtain prototype-guided similar region features. Then, PD^{2}Net extracts high-confidence and suspicious regions in the similar region features guided by the prototypes, in which the latter are mapped to shallow features for re-detection. Finally, the three-classification mask is generated by prototype guidance for high-confidence and re-detected regions.

were classified to distinguish the source from the tampered region. This serial approach is much better than the BusterNet parallel network, but the training process is not end-to-end. In addition, for copy-move forgery detection, the existing methods use the generator of the GAN network to perform the detection and distinguishment task of the source and tampered regions and optimize the detection and distinguishment result by the zero-sum game of generator and discriminator [57], [58], [59]. Among them, Islam et al. [60] proposed Doa-GAN in which the generator leverages second-order attention to first locate the approximate target region to exclude suspicious objects and then refine the detection results through global modeling. Zhang et al. proposed CNN-T GAN [61] to extract local and global features through CNN and Transformer, respectively, and introduce a Feature Coupling Layer between corresponding layers to integrate local and global information [62]. Jing et al. [63] proposed DMNet, which extracts tampered regions through decoupled edge supervision and decouples source domain features from the source and tampered domains. Finally, it optimizes the detection of similar regions by leveraging the source and tampered domains to achieve refined detection results. Hao et al. [64] proposed CMFDTR, which adopts the ViT-based structure as the encoder, processes the image in a sequence-to-sequence manner, and replaces the feature correlation calculation in the previous method with self-attention calculation. While the methods above achieve certain successes, they fail to obtain edge-structurally complete, object-level distinguishment results.

In contrast to the discriminative paradigm employed by existing approaches, this paper proposes a generative paradigm that addresses the challenge of distinguishing between the source region and the tampered region with identical data distributions from an unexplored perspective. Furthermore, the proposed method can refine detection targets and achieve object-level detection.

III. Method

This work aims to distinguish the source and tampered regions by introducing two different prototypes to represent the source and tampered regions in the copy-move forgery image. In this section, we introduce PD^{2}Net, which consists of the overall architecture of PD^{2}Net, a suspect region selection module that extracts warning regions in different similar features guided by the source and the tampered prototypes, and a suspect region re-identification module that obtained the supplementary target area. Fig. 3 shows an overview of the PD^{2}Net.

A. The Overall Architecture of PD^{2}Net

The input of PD^{2}Net is a copy-move forgery image. First, the input image is processed through an improved VGG16 network for feature extraction, and then the extracted features are input into the relevance extraction module to obtain coarse similar region feature $F_c \in \mathbb{R}^{H \times W \times C}$. We adopt VGG16 because copy-move forgeries require strong low-level texture modeling, for which VGG-style local features remain highly effective. Then, we initialize prototypes $P_t \in \mathbb{R}^{H \times W \times C}$ representing tampered regions and $P_s \in \mathbb{R}^{H \times W \times C}$ representing source regions according to the Gaussian distribution, by defining two objects to distinguish between them. We input the prototypes and F_c to the proposed Alternating Update (AU) module. During the network training process, the AU module facilitates the continuous updating of P_t and P_s, progressively enabling them to approximate the target tampered region and the source region. Additionally, it outputs two distinct similar region feature maps, denoted as F_{ct} and F_{cs}, both with dimensions $H \times W \times C$.

Then, we leverage F_{ct} and F_{cs} to address the limitations of traditional methods that often overlook non-significant regions in the presence of highly dominant high-confidence regions by a novel approach combining Suspicious Selection (SS) and

Suspicious Refinement (SR). This combined method ensures comprehensive detection by not only focusing on high-confidence regions but also identifying and refining potentially suspicious areas that may otherwise be suppressed or ignored.

First, the tampered feature map F_{ct} and the source feature map F_{cs} are input into the SS module to find inconsistent regions between the two. These inconsistent regions are denoted as $F_s \in \mathbb{R}^{H\times W\times C}$, characterized as areas that do not align well with the high-confidence regions derived from F_{ct} and F_{cs}. While these regions may appear insignificant compared to the consistent and dominant high-confidence regions, they often exhibit subtle patterns indicative of tampering. The role of SS is to isolate such regions for further examination, ensuring that no potentially critical information is lost during the detection process.

The feature maps F_{ct} and F_{cs} are first processed using ReLU activation functions to filter out background noise and retain only the significant features. This operation emphasizes regions with higher feature intensity, effectively narrowing the focus to areas of interest. After activation, an element-wise multiplication is performed between F_{ct} and F_{cs}, resulting in a high-confidence region map that highlights the areas where the tampered and source features strongly agree. This high-confidence map serves as the primary focus on reliable detection, while regions excluded from this map are flagged as suspicious and passed to the SR module for detailed refinement.

Then, F_s and the feature extracted from VGG16 are input into the SR module to handle these suspicious regions independently. By conducting fine-grained correlation detection, SR examines these regions to extract additional relevant features $F_{ac} \in \mathbb{R}^{H\times W\times C}$ that might represent subtle tampering artifacts or overlooked inconsistencies. This refinement process is essential for recovering valuable information that might not be immediately apparent in the high-confidence regions. The SR module thus complements the high-confidence regions by ensuring that even low-intensity or indistinct tampered areas are given appropriate attention.

By combining SS and SR, the proposed method effectively balances the focus between high-confidence and suspicious regions. This dual-stage approach avoids the common problem of overshadowing non-significant target regions by dominant features, which can lead to false negatives in tampering detection. Instead, it provides a comprehensive framework for analyzing both prominent and subtle features, significantly improving the robustness and accuracy of the model. The specific operations of the SS and SR modules are detailed in Sections III-E and III-F.

Finally, the high-confidence regions and the output feature F_{ac} of the SR module are input into the Prototype-guided distinguishment module to obtain the source and tampered region by corresponding to the updated prototypes.

B. Feature Extractor

We leverage the first five standard blocks of VGG16 as feature extractors for the input image to obtain feature F_v. We remove the pooling layer of the fourth and fifth blocks, respectively, and the standard convolution layer replaces the latter with dilated convolution. The dilation rate parameter of the dilated convolution module is r=1,2,4.

C. Correlation Extractor With Spatial Attention

The extracted feature F_v is performed self-correlation with spatial attention to generate the coarse similar region feature $F_c \in \mathbb{R}^{H\times W\times C}$.

To be specific, we use $F_v(i,j)$ represents the c-dimensional descriptor at (i,j). $F_v \in \mathbb{R}^{H\times W\times C}$, $i \in [1,h]$, $j \in [1,w]$, h and w represent the height and width of the feature after mapping, and in our work we let $h = w$. Prior to performing the attention and correlation calculations, an L2 normalization process is applied, $\bar{F}_v(i,j) = L2\_norm(F_v(i,j)) = F_v(i,j)/\|F_v(i,j)\|_2$. Applying L2 normalization allows us to standardize the value range of the feature representation, resulting in a normalized feature map $\bar{F}_v(i,j)$.

Spatial attention is a self-attention mechanism that computes the value of a specific position as a weighted sum of all positions. This approach captures long-range dependencies by assigning attention weights based on the similarity of colors and textures in the input features. In this paper, spatial attention is employed to enhance $\bar{F}_v$. First $\bar{F}_v$ is transformed into two feature spaces $f(\bar{F}_v) = \bar{F}_v W_{v,f} + b_{v,f}$ and $g(\bar{F}_v) = \bar{F}_v W_{v,g} + b_{v,g}$. Then we compute:

$$\beta_v^{(m,n)} = \frac{exp\left(s_v^{(m,n)}\right)}{\sum_n exp\left(s_v^{(m,n)}\right)} \tag{1}$$

where

$$s_v^{(m,n)} = f\left(\bar{F}_l^{(m)}\right)^T g\left(\bar{F}_v^{(n)}\right) \tag{2}$$

$\beta_v^{(m,n)}$ represents how much attention the model pays to the n position when predicting the m region, $m, n \in [1, h\times w]$. The output of the attention block is calculated as:

$$o_v^{(m)} = \sum_n \beta_v^{(m,n)} h\left(\bar{F}_l^{(n)}\right) \tag{3}$$

where $h(\bar{F}_v) = \bar{F}_v W_{v,h} + b_{v,h}$. Note that $W_{v,f}, W_{v,g} \in \mathbb{R}^{c\times\frac{c}{8}}$, $W_{v,h} \in \mathbb{R}^{c\times c}$, $b_{v,f}$, $b_{v,g} \in \mathbb{R}^{\frac{c}{8}}$, $b_{v,h} \in \mathbb{R}^c$, which are implemented as 1×1 convolutional layers. $O_v = \{o_v^{(1)}, o_v^{(2)}, \ldots, o_v^{(h\times w)},\}$ are the attention values for $\bar{F}_v$. Consequently, the spatial attention feature mapping through convolution can be expressed as:

$$\ddot{F}_v = Atten_v\left(\bar{F}_v\right) = \lambda_v O_v + \bar{F}_v \tag{4}$$

where λ_v is a learnable scaling parameter, initialized to 0.

Self-correlation aims to measure the similarity between two positions within a convolutional feature map, often utilizing scalar products for this purpose:

$$c_v^{(m,n)} = \left(\ddot{F}_v^{(m)}\right)^T \ddot{F}_v^{(n)} \tag{5}$$

Thus, we can obtain an initial correlation mapping tensor $C_v = \{c_v^{(m,n)} \mid m, n \in [1, h\times w]\} \in \mathbb{R}^{h\times w\times (h\times w)}$. In fact, among the features obtained, only a small fraction are highly relevant,

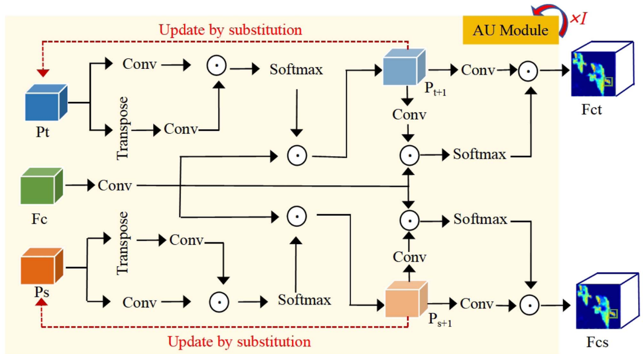

Fig. 4. The schematic illustrations of the AU module.

while the majority show little similarity. This suggests that the subset of C_v holds sufficient information to determine the appropriate feature for matching. Therefore, we sort C_v along the $(h \times w)$ channel and select the top-T value:

$$\tilde{C}_v\left(i, j, 1:T\right) = Top\_T\left(Sort\left(C_v\left(i, j, :\right)\right)\right) \tag{6}$$

Subsequently, a guiding and normalization operation is performed on $\tilde{C}_v$ to constrain the relevant values within a specific range and eliminate redundant values:

$$F_c = L2\_norm\left(Max\left(\tilde{C}_v, 0\right)\right) \tag{7}$$

D. Alternating Update Module

The AU module is designed to iteratively refine the source and tampered prototypes by aligning them with region-level features. At each iteration, the module alternates between estimating region-prototype associations and updating the prototypes accordingly. This mechanism ensures that the prototypes gradually converge toward representative patterns of the source and tampered regions.

As shown in Fig. 4, we utilize P_t and P_s as the query features, while F_c serves as both the key and value in the cross-correlation modeling (CCM) process. Through this mechanism, we iteratively update P_t and P_s to generate P_{t+1} and P_{t+1}, which are subsequently represented in the form of $\mathbb{R}^{H\times W\times C}$. This iterative update strategy is designed to progressively refine the feature representations, compelling P_t and P_s to move closer to their corresponding target regions: the tampered object for P_t and the source object for P_s. In this way, the model achieves improved discrimination between the tampered and source regions. The mathematical formulation of this process is expressed as follows:

$$\begin{aligned} P_{(t+1)} &= \text{Softmax}\left(\frac{conv(P_t)\cdot conv(F_c)^T}{\sqrt{D_{F_c}}}\right)\cdot conv(F_c),\\ P_{(s+1)} &= \text{Softmax}\left(\frac{conv(P_s)\cdot conv(F_c)^T}{\sqrt{D_{F_c}}}\right)\cdot conv(F_c), \end{aligned} \tag{8}$$

where the updated feature maps, $p_{(t+1)}$ and $p_{(s+1)}$, maintain the same spatial and channel dimensions as the original queries, specifically $H \times W \times C$. The operations within the equation include convolution ($conv(\cdot)$) applied to P_t, P_s, and F_c, followed by a dot product operation with the transposed key $conv(F_c)^T$. The resulting dot product values are normalized by the square root of the dimension of F_c (denoted as $\sqrt{D_{F_c}}$), ensuring stability in gradient computation and numerical representation.

The Softmax function transforms the raw correlation scores into a probability distribution that captures the relative importance of each spatial position in the key-value interaction. This weighted aggregation step allows the model to selectively focus on the most relevant features in F_c that contribute to refining the tampered and source regions.

After completing the iterative updates of the prototypes, we further leverage the optimized tampered region prototype P_{t+1} and source region prototype P_{s+1} to refine the coarse similar region features. This process is achieved through alternating embedding of the updated prototypes, enabling dynamic optimization to detect similar regions more accurately in the feature space. Specifically, we use P_{t+1} and P_{s+1} to guide the feature map F_c, generating similar region features associated with the tampered and source regions, respectively. By employing this alternating embedding strategy, the feature representations are dynamically adjusted, making the model more sensitive to similar regions. The mathematical formulation of this optimization process is as follows:

$$\begin{aligned} F_{ct} &= \text{Softmax}\left(\frac{conv(F_c)\cdot conv(P_{t+1})^T}{\sqrt{D_{P_{t+1}}}}\right)\cdot conv(P_{t+1}),\\ F_{cs} &= \text{Softmax}\left(\frac{conv(F_c)\cdot conv(P_{s+1})^T}{\sqrt{D_{P_{s+1}}}}\right)\cdot conv(P_{s+1}) \end{aligned} \tag{9}$$

where F_{ct} represents the similar region features guided by the updated tampered prototype P_{t+1}, and F_{cs} represents those guided by the updated source prototype P_{s+1}. F_{ct} and F_{cs} are in size of $H \times W \times C$. $\sqrt{D_{P_{t+1}}}$ and $\sqrt{D_{P_{s+1}}}$ are dimension normalization factors for the tampered and source prototypes, respectively, ensuring numerical stability during the dot product operation. The softmax function is applied to transform the raw correlation scores into a probability distribution, capturing the spatial relationship between the feature map F_c and the prototypes P_{t+1} and P_{s+1}. This weighted aggregation enables the model to extract features relevant to the tampered and source regions, significantly improving the accuracy of similar region detection.

The effectiveness of dynamic optimization is the alternating embedding of the two prototypes and the similar region feature F_c. Specifically, the F_c obtained through correlation detection enables P_t and P_s to focus on the tampered region and the source region, respectively, while establishing a clearer association between the two. In addition, the dynamically optimized F_{ct} and F_{cs} provide two feature descriptions of similar regions from different perspectives. These features effectively capture the details of tampered and source regions and can also be optimized by complementing each other to obtain a refined similar target.

We iteratively apply the AU module, with the number of iterations set to 4 in this paper, enabling gradual optimization of the prototype and similar region features.

E. Suspect Region Selection Module

The SS module identifies candidate regions that may correspond to copy-move pairs by computing multi-scale similarity between local features.

We define the suspect region as the location where a detection difference exists between F_{ct} and F_{cs}, i.e., where one is detected as background while the other is detected as a target. We extract the suspect region by computing the inconsistency between corresponding positions in two similar regions, feature F_{ct} and F_{cs}.

We perform multi-scale (with 1×1 and 2×2 sliding windows) and multi-dimensional (position-wise and channel-wise) extraction of the suspect region. Using 1×1 sliding window size, we calculate the inner product of corresponding positions to obtain a matrix M_1 representing consistent detection in F_{ct} and F_{cs} position-wise:

$$M_1 = F_{ct}(i,j) \cdot F_{cs}(i,j) \tag{10}$$

where i represents the i-th row, j represents the j-th column, and $M_1 \in \mathbb{R}^{H \times W \times 1}$

In addition, by computing the inner product of corresponding channels at corresponding positions, we obtain a matrix M_k representing consistency in F_{ct} and F_{cs} detection channel-wise:

$$M_k = F_{ct}^{\mathrm{k}}(i,j) \cdot F_{cs}^{\mathrm{k}}(i,j) \tag{11}$$

where k represents the k-th channel, and $M_k \in \mathbb{R}^{H \times W \times C}$.

We concatenate M_1 and M_k and perform convolution to obtain the consistent detection region M in F_{ct} and F_{cs}:

$$M = conv(M_1 || M_k) \tag{12}$$

where M is in size of $H \times W \times (C+1)$.

Similarly, using a 2×2 sliding window with stride 1, we obtain the consistency detection region $\hat{M}$ in F_{ct} and F_{cs}:

$$\begin{aligned} M_2 &= F_{ct}(v,w) \cdot F_{cs}(v,w), \\ \hat{M}_k &= F_{ct}^{\mathrm{k}}(v,w) \cdot F_{cs}^{\mathrm{k}}(v,w), \\ \hat{M} &= conv(M_2 || \hat{M}_k) \end{aligned} \tag{13}$$

where (v,w) represents the position after window partitioning, $M_2 \in \mathbb{R}^{(H-1)\times(W-1)\times 1}$, $\hat{M}_k \in \mathbb{R}^{(H-1)\times(W-1)\times c}$, $\hat{M} \in \mathbb{R}^{(H-1)\times(W-1)\times(C+1)}$. The detailed visual explanation is shown in Fig. 5.

We reshape $\hat{M}$ to $H \times W \times (C+1))$ and merge it with M to obtain the final consistent detection region in F_{ct} and F_{cs} at multiple scales. Subsequently, we use a reverse gate to obtain the suspect region, represented as F_s:

$$F_s = 1 - Sigmoid(conv(Reshape(\hat{M}) + M)) \tag{14}$$

F. Suspect Region Re-Identification Module

The SR module re-evaluates and filters the candidate region pairs generated by SS to produce accurate detection. It leverages prototype-guided similarity reasoning to suppress false matches and enhance true copy-move pairs. This refinement step provides the final localization masks for both the source and tampered regions.

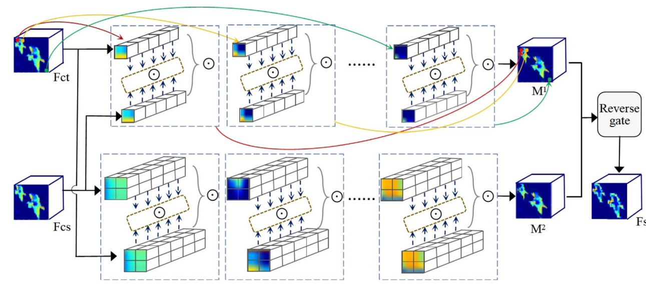

Fig. 5. The schematic illustrations of the SS module.

We re-identify the suspect regions obtained through the SS module to eliminate false positive areas and extract target region features F_a that are not included in the high-confidence regions. F_a serves as a supplement to the high-confidence regions, enabling the generation of refined similar region pairs.

We utilize the features from the last layers of the fourth and fifth blocks of the feature extractor for suspect region re-identification. Specifically, we map the suspect regions in F_s to the features F_4 and F_5 to locate the suspect regions within F_4 and F_5. We retain the elements in F_4 and F_5 corresponding to the suspect positions while masking other elements, including the background and high-confidence regions. The masked F_4 and F_5 are defined as F_{4m} and F_{5m}. The formulas for the above operations are as follows:

$$\begin{aligned} F_{4m} &= \mathrm{Position}(F_s) \rightarrow \mathrm{Mapping}(F_4), \\ F_{5m} &= \mathrm{Position}(F_s) \rightarrow \mathrm{Mapping}(F_5), \end{aligned} \tag{15}$$

Then, we perform correlation extraction on the masked features from F_{4m} and F_{5m}, obtaining features F_{4c} and F_{5c}, respectively. Finally, we adaptively fuse F_{4c} and F_{5c} to generate F_{ac}. The formula for the operation is given below.

$$F_{ac} = Fuse(Corr(F_{4m}), Corr(F_{5m})) \tag{16}$$

The mapping process is essentially a decoupling operation that eliminates the influence of non-target backgrounds and high-confidence target regions. This ensures that correlation extraction focuses the network on the suspect regions, thereby addressing the issue of misjudgment caused by suspect regions being masked by high-confidence regions. Furthermore, selecting F_4 and F_5 for mapping is based on the fact that shallow features retain effective detail information, which is often lost in deeper layers of the network. Therefore, re-identifying similar regions in F_4 and F_5 and using them as a supplement to the high-confidence regions not only reduces false positives but also enhances the detection of target regions.

G. Loss Function

The network parameters are optimized during the training process using both prototype and classification loss. The overall loss function is shown below:

$$\mathcal{L} = \alpha\mathcal{L}_{\mathrm{sp}} + \beta\mathcal{L}_{\mathrm{tp}} + (1-\alpha-\beta)\mathcal{L}_{\mathrm{c}} \tag{17}$$

where $\mathcal{L}_{\mathrm{sp}}$ and $\mathcal{L}_{\mathrm{tp}}$ denote the source and the tempered region prototypes loss, respectively. $\mathcal{L}_{\mathrm{c}}$ represents the three-classification loss, including background, source, and tampered region. The α, β, and $1-\alpha-\beta$ are modeled as learnable parameters and jointly optimized with the network via back-propagation. To ensure valid weighting, the three loss weights corresponding to α, β, and $1-\alpha-\beta$ are normalized using a softmax function. This guarantees that all three weights are non-negative, bounded within (0,1), and sum to one, thereby providing a stable weighting mechanism for balancing different loss components.

We use Focal loss [65] to compute $\mathcal{L}_{\mathrm{sp}}$ and $\mathcal{L}_{\mathrm{tp}}$, aiming to mitigate the negative impact of foreground and background imbalance in the prototype representations.

We calculate $\mathcal{L}_c$ using spatial cross-entropy, with the formula as follows:

$$\begin{aligned}\mathcal{L}_{\mathrm{c}} = &-\frac{1}{HW}\sum_{i=1}^{W\times H} w_1{\cdot}\hat{p}_i \cdot \log(p_i)\\ &+ w_2 \cdot (1-\hat{p}_i)\cdot \log(1-p_i)\end{aligned} \tag{18}$$

where $p_i \in \{0,1\}$ indicates whether the i-th pixel is part of the target regions under consideration, while $\hat{p}_i$ represents the predicted probability of this pixel being classified correctly. The w_1 and w_2 are introduced to adjust the influence, with values set at 0.8 and 0.2, respectively.

IV. Experiments

A. Training Details

We employ the PyTorch framework [66] to develop and train PD²Net. Following approximately 25 epochs of training, the network exhibited signs of convergence. For optimization, we utilized the Adam optimizer with a learning rate set to $1e-2$ and a batch size of 16 [67]. To expedite network optimization and reduce parameters, we resized the dataset images to 256×256 dimensions for training. All experiments are conducted using a single NVIDIA 16 GB Tesla V100 GPU.

The proposed PD²Net consists of approximately 158.37 million parameters. Under the default training setting, the average training time per epoch is about 28 minutes, while the inference time for a single image is 0.16 seconds. To mitigate the increased computational burden introduced by the large parameter count, we employ a parallel training strategy. Compared to the traditional training scheme, this design significantly reduces the training overhead and improves overall efficiency. Compared to the traditional training scheme, which takes approximately 69 minutes per epoch, the parallel computing strategy achieves a substantial reduction of 41 minutes in epoch-level training time.

B. Dataset

We utilize the USC-ISI dataset constructed by Wu et al. [55] for model training, validation, and test sets at a ratio of 8:1:1. The USC-ISI dataset comprises 100,000 samples, which serves as a large-scale synthetic dataset ensuring diverse coverage of scenarios.

In addition to the USC-ISI training set, we employed two additional datasets, CASIA v2.0 [68] and CoMoFoD [69], to systematically evaluate the generalization of PD²Net. Both datasets introduce substantial domain shifts in image content, tampering patterns, and degradation levels, which pose significant challenges for forgery localization models. By testing on these datasets, we aim to examine how the prototype-guided representation learning in PD²Net adapts to unseen data distributions and preserves discriminative consistency between source and tampered regions under varying attack conditions.

Specifically, CASIA v2.0 consists of 7491 genuine samples and 5123 tampered samples, encompassing copy-move forgery methods as well as other techniques like cropping and deletion. We selected 1313 copy-move samples for analysis according to the specific tasks.

CoMoFoD includes 200 original copy-move images. Each image has undergone 25 types of post-processing operations, including rotation, scaling, blurring, etc., resulting in a total of 5,000 images. Therefore, we use this dataset to test the robustness of the PD²Net.

C. Evaluation Metrics

To assess the effectiveness of PD²Net, we employ precision, recall, and F-1 score as metrics to evaluate the performance of PD²Net at the pixel level. The formulas of precision and recall are shown as follows:

$$Precision = \frac{TP}{TP+FP} \tag{19}$$

$$Recall = \frac{TP}{TP+FN} \tag{20}$$

where, TP, FP, and FN represent true positive, false positive, and false negative detection regions.

We employ the F1 to comprehensively assess PD²Net, representing the harmonic mean of precision and recall, which is shown as follows:

$$F1 = 2\cdot\frac{Precision \cdot Recall}{Precision + Recall} \tag{21}$$

D. Contrast Methods

To validate the superiority of our proposed PD²Net, we selected four paradigms of advanced deep learning-based copy-move forgery image detection and distinguishment for comparison, including tampered region detection-based (BusterNet [55]), classification-based (CMSDNet [56]), GAN-based (Doa-GAN [60] and CNN-T GAN [61]) paradigms, and ViT-based (CMFDTR [64] and LBRT [54]) methods.

a) BusterNet [55]: Fuse the similar regions and tampered regions features to achieve source and tampered region detection and distinguishment.

b) CMSDNet [56]: Uses multi-scale feature extraction to identify similar regions and crop them out in the image to classify them.

TABLE I
RESULTS OF PRECISION, RECALL, F1 SCORE (%) ON PRISTINE, SOURCE, AND TARGET REGIONS OF CONTRAST APPROACHES AND PD^{2}NET ON USC-ISI, CASIA v2.0, AND COMOFOD DATASETS.

Datasets	Methods	Precision			Recall			F1		
		Pristine	Source	Tamper	Pristine	Source	Tamper	Pristine	Source	Tamper
USC-ISI	BusterNet	91.01	23.83	10.19	98.72	11.44	2.91	94.51	12.84	3.99
	CMSDNet	97.02	50.68	26.24	97.84	45.48	43.32	97.37	45.35	30.40
	Doa-GAN	96.68	68.65	74.64	98.33	58.89	78.65	97.46	60.29	74.79
	CNN-T GAN	98.71	86.12	87.10	99.14	84.37	87.44	98.92	84.45	86.36
	LBRT	97.03	86.09	85.53	99.07	78.93	80.21	97.92	74.35	84.81
	CMFDTR	96.86	**86.86**	84.39	98.64	69.62	81.55	97.74	77.29	82.94
	PD^{2}Net	**99.46**	85.04	**88.39**	**99.57**	**85.62**	**88.42**	**98.97**	**86.51**	**86.69**
CASIA	BusterNet	94.45	20.93	6.36	98.01	22.51	1.64	96.00	18.58	2.14
	CMSDNet	93.57	8.66	48.53	99.22	2.28	28.43	96.08	2.97	30.58
	Doa-GAN	96.43	31.88	22.86	97.39	36.74	26.91	96.75	28.86	22.71
	CNN-T GAN	96.19	32.65	25.85	99.14	39.83	28.64	97.35	29.72	25.32
	LBRT	95.78	32.45	40.91	97.01	35.37	26.38	96.93	26.72	29.51
	CMFDTR	93.40	31.73	**55.97**	98.51	16.39	7.59	95.89	21.61	13.37
	PD^{2}Net	**96.78**	**35.29**	43.48	**99.91**	**41.45**	**29.20**	**98.39**	**30.98**	**30.87**
CoMoFoD	BusterNet	96.92	15.88	5.81	94.87	25.03	1.64	95.64	14.38	2.18
	CMSDNet	97.95	32.05	27.52	98.35	29.53	29.63	98.13	28.15	26.58
	Doa-GAN	94.22	15.47	12.60	**99.63**	6.09	4.71	96.69	7.25	5.63
	CNN-T GAN	98.28	29.79	22.63	98.25	44.74	46.27	97.72	28.63	21.36
	LBRT	98.02	35.94	27.81	97.43	42.68	45.82	98.06	29.73	30.69
	PD^{2}Net	**98.98**	**38.59**	**28.93**	99.62	**57.13**	**51.05**	**98.36**	**31.64**	**32.08**

c) Doa-GAN [60]: Employs a GAN structure for source and tampered region distinguishment, where the generator consists of a dual-attention-based feature extractor.

d) CNN-T GAN [61]: Employs a GAN structure for source and tampered region distinguishment, where the generator employs both CNN and Transformer simultaneously and conducts information interaction between the two.

e) CMFDTR [64]: Employs ViT structure as the encoder and achieves information integration through a top-down pathway in the decoder.

f) LBRT [54]: Performs ViT encoding on the global blocks segmented from the image and the local blocks re-segmented, respectively. Fuses two branches to realize distinguishment.

Additionally, to assess the effectiveness of our proposed prototype-guided generative detection and discrimination mechanism from a paradigmatic perspective, we also selected binary classification methods used solely for copy-move forgery image detection, including ARNet [52], DRNet [53], and CAMU [51], for comparative experiments. By embedding binary classification methods into different paradigms used in distinguishment to enrich the comparison methods, we further demonstrated the superiority of the generative PD^{2}Net.

g) ARNet [52]: Extracts relevant features through multi-scale modeling, then optimizes through an encoder-decoder to detect without distinguishing between source and tampered regions.

h) DRNet [53]: Detect similar regions by calculating the self-correlation on the decoupled shallow features and the last layer's features.

i) CAMUNet [51]: Detect similar regions utilizing coordinate attention for multi-scale feature extraction, followed by fusion through upsampling.

E. Comparison With State-of-the-Arts

In this section, we compare the methods directly used for detection and distinguishment, including BusterNet, CMSDNet, Doa-GAN, and CNN-T GAN, with PD^{2}Net to verify the validity of our proposed method.

Table I clearly demonstrates that our PD^{2}Net outperforms the other methods, providing more accurate distinguishment between source and tampered regions. On the USC-ISI dataset, compared to the sub-optimal CNN-T GAN, our PD^{2}Net improves the F1 for detecting the source region by 2.06%. Additionally, to verify the generalization of our proposed PD^{2}Net, we conducted experiments on the CASIA and CoMoFod datasets, where our method still achieves overall optimal results. On the CASIA dataset, except that the precision of tampered region detection is lower than that of CMFDTR, which is based on ViT, all other indicators are optimal. In particular, compared with CNN-T GAN, the precision of this method in detecting the source region has increased by 2.64%. In terms of the F1 metrics in the source and the tampered region, compared with the LBRT method based on ViT, the proposed method has improved by 4.26% and 1.36% respectively. On the CoMoFoD dataset, compared with LBRT, the F1 score of this method in source region detection has increased by 1.91%.

Fig. 6 illustrates visualizations of masks obtained by the comparative methods and PD^{2}Net. We can observe that PD^{2}Net can detect and locate structurally intact source and tampered region objects. Specifically, in the second row of Fig. 6, the masks detected by the comparative networks have smoother edges of the flowers and lack details. In contrast, our PD^{2}Net produces more refined detection results closer to the ground truth, and we can clearly see the outline of the petal.

F. Analysis of Generative Paradigm

We incorporate binary classification methods that focus solely on detection without distinguishment as plugins, embedding them into three existing discriminative copy-move forgery detection and distinguishment paradigms as comparative experiments to validate the effectiveness of PD^{2}Net of generative paradigm.

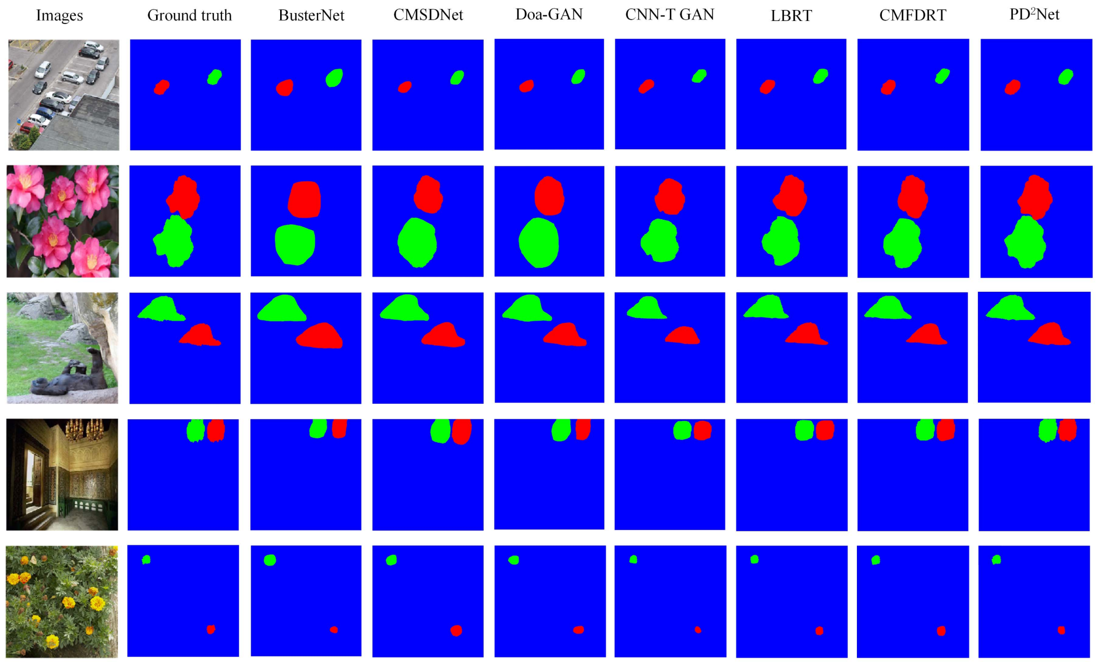

Fig. 6. Compare experimental visualizations, where the red is the tampered region, the green is the source region, and the blue is the pristine region.

TABLE II
RESULTS OF F1 (%) ON PRISTINE, SOURCE, AND TARGET REGIONS OF CONTRAST APPROACHES, OBTAINED BY EMBEDDING BINARY CLASSIFICATION METHODS INTO THREE PARADIGMS, AND PD^{2}NET ON USC-ISI, CASIA V2.0, AND COMOFOD DATASETS.

Datasets	Paradigms	ARNet			DRNet			CAMUNet		
		Pristine	Source	Tamper	Pristine	Source	Tamper	Pristine	Source	Tamper
USC-ISI	Original	94.11	23.62	29.13	94.03	31.63	30.72	95.01	29.62	31.26
	Temper region-based	97.87	30.66	37.18	97.10	39.81	57.49	97.39	36.98	59.71
	Classification-based	97.73	69.12	74.84	97.32	73.77	76.83	97.81	70.04	74.69
	GAN-based	98.66	82.63	80.18	98.46	85.16	82.32	98.69	82.47	81.15
	PD^{2}Net	**98.97**	**86.51**	**86.69**	**98.97**	**86.51**	**86.69**	**98.97**	**86.51**	**86.69**
CASIA	Original	90.96	9.33	8.97	92.73	8.69	13.83	91.17	8.06	10.98
	Temper region-based	96.89	13.63	13.87	96.33	13.12	17.62	96.53	13.49	15.83
	Classification-based	98.36	23.56	19.26	98.69	25.36	28.26	98.33	22.76	20.94
	GAN-based	98.25	27.15	23.59	98.24	29.28	25.01	98.02	28.04	21.86
	PD^{2}Net	**98.39**	**30.98**	**30.87**	**98.39**	**30.98**	**30.87**	**98.39**	**30.98**	**30.87**
CoMoFod	Original	90.75	8.74	10.62	93.82	11.85	10.73	90.62	6.13	9.83
	Temper region-based	96.84	15.89	17.73	96.58	13.08	15.24	96.04	11.63	13.86
	Classification-based	98.16	21.84	21.84	98.12	23.75	24.82	98.01	20.46	22.84
	GAN-based	97.65	19.15	23.59	97.24	29.28	25.01	97.62	22.04	26.86
	PD^{2}Net	**98.36**	**31.64**	**32.08**	**98.36**	**31.64**	**32.08**	**98.36**	**31.64**	**32.08**

Table II presents the F1 scores obtained by PD^{2}Net, directly using ARNet, DRNet, and CAMUNet and embedding them separately into tampered region-based, classification-based, and GAN-based approaches for copy-move forgery image detection and distinguishment. The overall metrics obtained by PD^{2}Net are higher than those of other methods, which verifies the effectiveness of our methods. We introduce prototypes, which are attributed to the definition of distinct prototypes for source and tampered regions, thereby improving discriminative capabilities by adding prior constraints. In addition, we observe that directly applying binary classification methods for distinguishment yields poor results. This is because binary classification methods inherently detect similar regions and lack the constraints to distinguish tampered and source regions with the same data distribution.

Inserting binary classification methods into three discriminative paradigms improves distinguishment results to some extent. This is because utilizing tampered region features as auxiliary information adds clues for distinguishing between the source and the tampered. The classification-based approach distinguishes between the source and the tampered region based on the binary mask. The superior binary detection results inevitably produce satisfying distinguishment results. In addition, generative adversarial models learn the true data distribution through a zero-sum game, enhancing their ability to detect and distinguish copy-move forgery images.

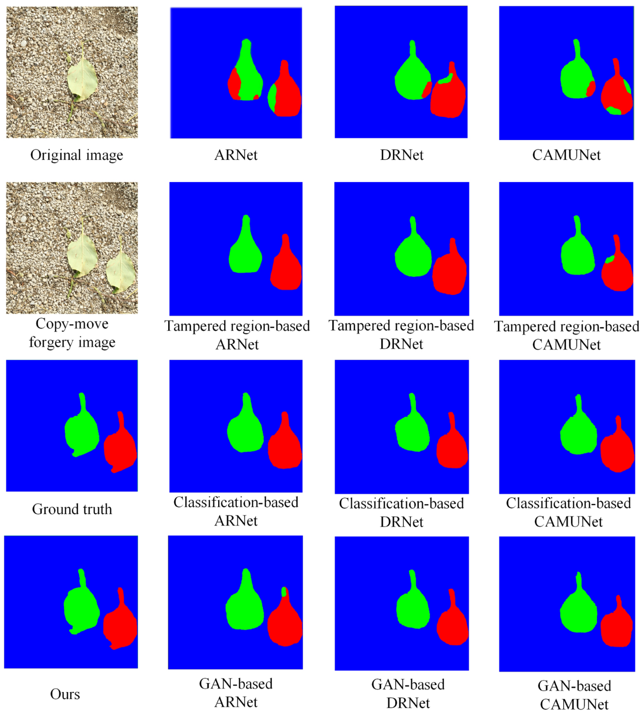

Fig. 7. The mask generated by embedding the binary classification methods into the existing three paradigms.

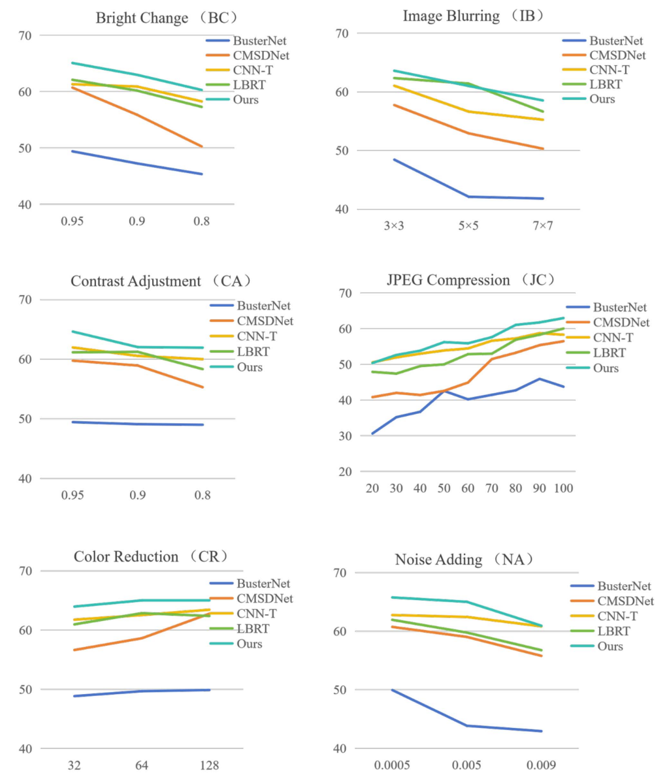

Fig. 8. Comparison of pixel-level F1 (y-axis) obtained by ours and other contrast methods on the images under different attacks (x-axis) from the CoMoFoD dataset.

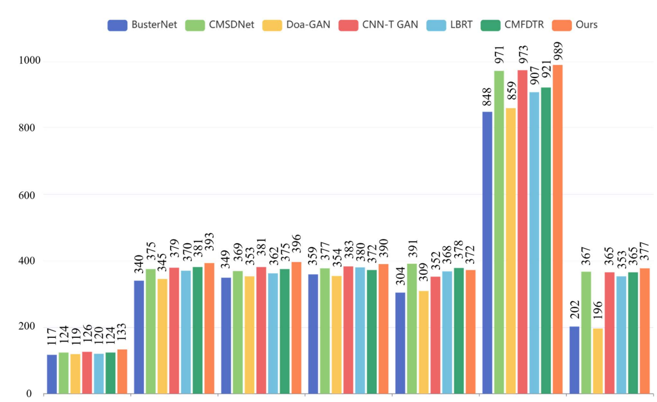

Fig. 9. The number of images under each attack correctly detected through CANet and other comparison networks.

However, the above three discriminative paradigms still fall short of our proposed PD^{2}Net. Specifically, our method achieves 6.49%, 4.37%, and 5.54% improvement in the F1 score for detecting the source and the tampered regions on the USC-ISI dataset compared to the GAN-based ARNet, DRNet, and CAMUNet, respectively. In addition, the proposed method improves the F1 score for detecting tampered regions on the CASIA dataset by 11.61%, 2.61%, and 9.93% compared to the classification-based ARNet, DRNet, and CAMUNet, respectively.

We present visual results, shown in Fig. 7, and analyze them to demonstrate the superiority of PD^{2}Net. The mask in the first row demonstrates the effectiveness of directly applying binary classification methods for distinguishment, where the source and tampered regions are mixed. Embedding binary classification methods into tampered region-based and GAN-based distinguishment paradigms leads to some improvement in performance, but confusion still exists in source and tampered regions. In contrast, the classification-based approach can effectively distinguishment between the two. However, it fails to obtain a complete contour of the detected object and is not end-to-end, adding complexity to the application. Additionally, as shown in Fig. 7, we observe that its detection results lack contour detail, whereas the mask generated by our method reveals the tips of the leaves.

G. Robustness Analysis

Copy-move forgery images are typically subjected to attack operations to conceal tampering traces. Therefore, the strong robustness of the proposed approach is necessary. We conduct robustness analysis on the CoMoFoD dataset, in which images are attacked in six ways [69], including JPEG compression (JC), Noise Adding (NA), Image Blurring (IB), Brightness Change (BC), Color Reduction (CR), Contrast adjustments (CA). The detailed parameters for each attack are shown in Table III. The Fig. 8 illustrates the F1 scores of the comparative methods and our proposed PD^{2}Net under the six attacks through a line chart. Except under the 5× 5 IB attack and JPEG compression with a quality factor of 20, the proposed method achieves the highest detection F1 score among all other types of attacks. Fig. 9 shows the number of images under each attack correctly detected through PD^{2}Net and other comparison methods, and we can find

TABLE III
PARAMETER DESCRIPTION OF ATTACK METHOD IN COMOFOD DATASET

Attack operation	**Parameter**
JPEG compression (JC)	Compression quality factors = {20, 30, 40, 50, 60, 70, 80, 90, 100}
Noise Adding (NA)	$\mu = 0, \sigma^2 = \{0.009, 0.005, 0.0005\}$
Image Blurring (IB)	mean filter = {3×3, 5×5, 7×7}
Brightness Change (BC)	(Lower bound, Upper bound) = {(0.01, 0.95), (0.01, 0.9), (0.01, 0.8)}
Color Reduction (CR)	Color reduction intensity levels = {32, 64, 128}
Contrast adjustments (CA)	(Lower bound, Upper bound) = {(0.01, 0.95), (0.01, 0.9), (0.01, 0.8)}

TABLE IV
PD^{2}NET RESULTS OF PRECISION, RECALL, F1 (%) ON PRISTINE, SOURCE, AND TARGET REGIONS OF ABLATION APPROACHES ON USC-ISI, CASIA V2.0, AND COMOFOD DATASETS.

Datasets	Methods	Precision			Recall			F1		
		Pristine	Source	Tamper	Pristine	Source	Tamper	Pristine	Source	Tamper
USC-ISI	Baseline	97.46	73.19	79.53	98.11	79.36	80.69	97.84	79.72	79.57
	Fc→prototype	97.97	78.73	83.53	99.10	80.34	84.81	97.90	83.57	82.45
	Updated prototype→Fc	98.93	81.83	85.62	99.24	84.54	86.07	98.31	84.61	84.64
	Fusion without mapping	98.92	83.12	86.09	99.56	84.97	86.34	98.48	85.89	85.62
	Mapping and fusion	**99.46**	**85.04**	**88.39**	**99.57**	**85.62**	**88.42**	**98.97**	**86.51**	**86.69**
CASIA	Baseline	96.60	26.36	35.69	99.40	33.83	20.76	96.46	21.64	19.98
	Fc→prototype	96.74	30.52	37.53	99.35	37.85	23.34	96.74	24.60	23.86
	Updated prototype→Fc	96.36	32.01	39.95	99.78	39.07	25.75	97.63	27.53	26.85
	Fusion without mapping	96.61	32.67	40.24	99.85	39.18	26.23	98.47	28.02	27.11
	Mapping and fusion	**96.78**	**35.29**	**43.48**	**99.91**	**41.45**	**29.20**	**98.39**	**30.98**	**30.87**
CoMoFod	Baseline	97.52	25.67	24.72	98.08	48.52	41.84	97.75	22.53	25.05
	Fc→prototype	97.94	29.39	26.07	99.21	52.76	44.90	97.59	25.84	27.68
	Updated prototype→Fc	98.52	35.43	27.58	99.35	55.01	47.76	98.06	28.01	30.03
	Fusion without mapping	98.61	36.04	27.63	**99.69**	55.84	49.02	98.09	29.25	31.01
	Mapping and fusion	**98.98**	**38.59**	**28.93**	99.62	**57.13**	**51.05**	**98.36**	**31.64**	**32.08**

that PD^{2}Net has the best performance. Under the NA attack, the proposed method correctly detected 12 more images than CMFDTR.

H. Ablation Experiment

To validate the effectiveness of each module proposed, we conducted ablation experiments.

a) Baseline: Introduce prototypes and concatenate them with the similar region feature.

b) Fc → prototype: Introducing and updating prototypes by CCM operations, with loss supervised for the prototypes.

c) Updated Prototype → Fc: Performing CCM operations on the updated prototypes separately and detecting and distinguishing by fusing the obtained features adaptively.

d) Fusion without mapping: Add the SS module and directly fuse the output of the SS module and the highly confident regions.

e) Mapping and fusion: Introduce SR and adaptively fuse the output of the SR module with the high-confidence region feature.

In the ablation experiments, the distinguishment of background, source, and tampered region is achieved by the prototype guidance.

Table IV presents the results of ablation experiments that verify the effectiveness of each module we proposed. Cross-correlation modeling between prototypes and coarse similar regions improves the distinguishment performance. Through iterative optimization, prototypes and coarse similar regions progressively approach the target.

We respectively present the heat maps of the prototypes during the network iteration process to verify their object-level prior capabilities in capturing the source domain and the tampered domain. As shown in the heat map visualization in Fig. 10, the evolution trend of the prototype during the training process is presented from left to right. In the early stage of training, the attention chart shows scattered and noisy activations, and the response areas often appear in irrelevant background areas. As the training progresses, the network gradually learns to suppress these ineffective activations. The response area with high confidence becomes more concentrated and clear, and can be aligned more accurately to the source or tampered objects. The above changing trends indicate that the network gradually optimizes its feature representation ability, forcing the prototype to gradually approach the target region.

The addition of the SS module does not obtain significant improvement, as it divides highly consistent and inconsistent detection results in the two representations of similar regions guided by different prototypes, which are then merged without further processing of suspicious regions, resulting in minimal impact.

However, the detection performance is further improved after adding the SR module to re-discriminate suspicious regions and

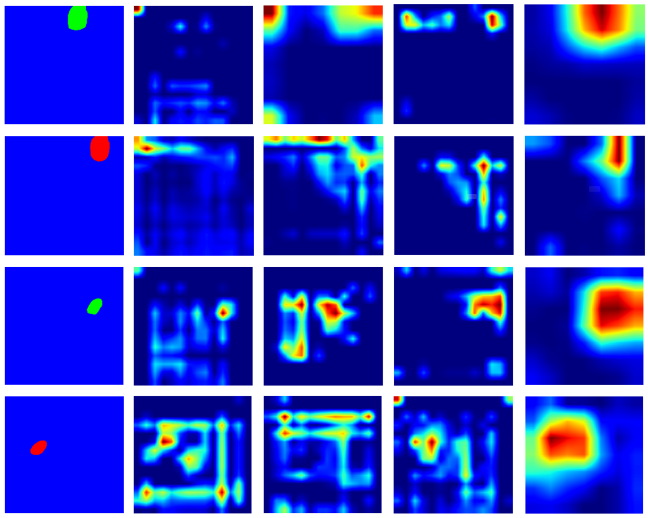

Fig. 10. Grad-CAM-based jet heatmaps illustrating the evolution of the source and tampered prototypes during training. The first column shows the ground-truth mask, and the subsequent columns visualize how the prototype responses become progressively focused and discriminative as the AU module iteratively updates them.

TABLE V
ABLATION STUDY ON LOSS FUNTION (F1 SCORE %). WE COMPARE FOCAL LOSS WITH STANDARD CROSS-ENTROPY LOSS UNDER THE SAME TRAINING SETTION.

Dataset	Loss Function	Source	Tampered	Average
USC-ISI	Cross-Entropy	85.72	85.94	85.83
	Focal Loss	86.51	86.69	86.60
CASIA	Cross-Entropy	29.41	29.03	29.22
	Focal Loss	30.98	30.87	30.93
CoMoFoD	Cross-Entropy	30.12	30.45	30.29
	Focal Loss	31.64	32.08	31.86

merging them with high-confidence regions. This is because we identified suspicious regions and mapped them to shallow features, thereby filtering out other regions and re-detecting only suspicious regions. This avoids the suppression of suspicious regions by high-confidence regions, thereby addressing the misidentification of suspicious regions. By using the SS module, we obtain the supplementary areas for refining the detection targets.

To further validate the effectiveness of Focal Loss, we conduct an ablation study by replacing it with standard Cross-Entropy Loss under the same training settings. The results are shown in Table V.

As observed, replacing Focal Loss with Cross-Entropy leads to consistent performance degradation across all datasets. The performance drop is more noticeable on CASIA and CoMoFoD, which contain more challenging cases such as small or low-contrast tampered regions.

This is because Cross-Entropy treats all samples equally, causing the optimization to be dominated by easy regions, while Focal Loss down-weights easy samples and emphasizes hard regions. This is particularly important in our framework, where accurate role assignment relies on refining ambiguous regions through prototype–feature interaction.

TABLE VI
PERFORMANCE OF THE F1(%) METRIC ON THE USC-ISI DATASET BASED ON THE BASELINE UNDER THE DIFFERENT T

	Prinstine	Source	Tamper
T=64	97.23	75.98	76.63
T=128	**97.84**	**79.72**	79.57
T=256	97.79	78.35	**80.91**

TABLE VII
EXPERIMENTAL RESULTS UNDER DIFFERENT NUMBERS OF CCM EXECUTIONS (F1 SCORE %)

Datasets	Methods	F1		
		Pristine	Source	Tamper
USC-ISI	AU (1)	97.90	83.57	82.45
	AU (2)	**98.36**	84.03	83.86
	AU (4)	98.31	**84.61**	**84.64**
	AU (6)	98.25	84.15	84.19
CASIA	AU (1)	96.74	24.60	23.86
	AU (2)	97.16	26.27	25.98
	AU (4)	**97.63**	**27.53**	**26.85**
	AU (6)	97.29	27.29	26.46

Moreover, we observe that the training process remains stable when using Focal Loss, without introducing optimization instability. These results confirm that Focal Loss contributes to both improved distinguishment performance and stable training in PD^{2}Net.

I. Hyperparameter Experimental Verification

1) The Selection of Top-T: The selection of the Top-T parameter is crucial to the overall detection performance. When T is excessively large, redundant and irrelevant information tends to be retained; conversely, when T is too small, some meaningful matches may be discarded. To determine an appropriate balance, we conducted experiments with T=64, 128, and 256, and finally adopted T=128 as it achieved the best trade-off between accuracy and computational efficiency. The corresponding experimental verification on the baseline is shown in Table VI.

2) Number of AU Iterations: We experiment by updating the source and the tampered prototype after 1, 2, 4, and 6 AU module runs respectively. The results in Table VII show that increasing the number of AU iterations from 1 to 4 consistently improves performance, demonstrating the effectiveness of iterative prototype refinement. Further increasing the number of iterations does not lead to additional gains and may introduce slight fluctuations. Therefore, 4 iterations are adopted as a practical setting. AU (1) and AU (4) correspond to the Fc $\rightarrow$ prototype and the Updated prototype $\rightarrow$ Fc in the ablation experiment, respectively.

J. Prototype Initialization Strategy Analysis

To validate the effectiveness of our initialization strategy, we compare Gaussian initialization with random uniform initialization, zero initialization, and pre-trained feature centroids on the USC-ISI dataset. As shown in Table VIII, Gaussian initialization

TABLE VIII
EXPERIMENTAL RESULTS UNDER DIFFERENT PROTOTYPE INITIALIZATION STRATEGIES (F1 SCORE %)

Datasets	Methods	F1		
		Source	Tampered	Average
USC-ISI	Zero Vector	80.83	81.59	81.71
	Random Uniform	83.24	85.29	84.27
	Feature centroids	86.34	86.81	86.58
	Gaussian (Ours)	86.51	86.69	86.60
CASIA	Zero Vector	25.86	32.84	29.35
	Random Uniform	28.98	29.42	29.20
	Feature centroids	23.39	27.12	25.26
	Gaussian (Ours)	30.98	30.87	30.93

TABLE IX
SENSITIVITY ANALYSIS UNDER DIFFERENT RANDOM SEEDS (F1 %)

Seed	Source	Tampered	Average
0	86.47	86.63	86.55
42	86.51	86.69	86.60
123	86.44	86.58	86.51

achieves the highest F1 scores (86.51% for source and 86.69% for tampered regions), outperforming uniform initialization by 2.53% and zero initialization by 5.59% on average. Zero-vector initialization forces all prototypes to start from identical representations, leading to indistinguishable gradients and a slow, unstable convergence process. Random uniform initialization suffers from the influence of extreme values, which introduces noise in the early optimization stages.

In addition, we further evaluate the use of pre-trained feature centroids, which strategy that embeds a strong dataset-dependent bias into the initial prototype space. As shown in Table VIII, centroid initialization shows inferior generalization performance compared with Gaussian initialization, particularly on cross-dataset evaluations. We attribute this to the fact that centroids reflect dominant patterns of the pre-training dataset and may suppress subtle variations associated with tampered regions, thereby limiting the flexibility of the Alternating Update mechanism to refine prototypes.

Overall, Gaussian initialization provides a neutral, unbiased starting point with moderate variance, enabling prototypes to adapt more effectively during iterative optimization. This behavior aligns with findings in prototype-learning literature, where Gaussian initialization is widely adopted for its stability and its ability to preserve symmetry before learning meaningful feature distinguishment.

To evaluate the sensitivity of the proposed method to stochastic factors, we conduct additional experiments using different random seeds. As shown in Table IX, PD^{2}Net achieves highly consistent F1 performance across multiple runs, with only negligible variance. This demonstrates that the proposed framework is robust to random initialization and stochastic optimization, ensuring stable performance and good reproducibility.

K. Cost Calculation Explanation

We provide detailed comparisons of model parameters (Param), inference speed (TPS, images per second), and computational cost (FLOPs), as summarized in Table X. From a trade-off perspective, PD^{2}Net demonstrates a more favorable balance between efficiency and effectiveness compared with existing methods.

First, compared with the lightweight model LBRT, PD^{2}Net has a larger number of parameters. However, it achieves superior efficiency and performance simultaneously. Specifically, PD^{2}Net achieves higher throughput (6.25 vs. 6.21 images/s) and maintains a moderate computational cost (42.96 G FLOPs), while significantly improving the F1 score across all datasets (86.60% vs. 79.58% on USC-ISI, 30.93% vs. 28.12% on CASIA, and 31.86% vs. 30.21% on CoMoFoD). This indicates that the increased model capacity is effectively utilized, leading to better accuracy without sacrificing efficiency.

Second, compared with CNN-T GAN, which achieves strong performance among existing methods, PD^{2}Net not only attains higher detection accuracy but also demonstrates substantially better efficiency. Specifically, PD^{2}Net achieves higher F1 scores (86.60% vs. 85.41% on USC-ISI, 30.93% vs. 27.52% on CASIA, and 31.86% vs. 25.00% on CoMoFoD), while using fewer parameters (158.37 M vs. 208.32 M), achieving higher throughput (6.25 vs. 2.04 images/s), and requiring lower computational cost (42.96 G vs. 64.78 G FLOPs).

These results demonstrate that PD^{2}Net does not simply rely on increased model complexity to improve performance. Instead, it achieves consistent gains in both accuracy and efficiency through the proposed prototype-guided assignment mechanism. Therefore, PD^{2}Net achieves a more favorable performance-efficiency trade-off compared with both lightweight and high-capacity baselines.

L. Scalability and Prototype Generalization Analysis

To evaluate the scalability of PD^{2}Net beyond the default 256×256 setting, we conduct additional inference experiments on higher-resolution inputs, including 512×512 and 1024×1024 images which obtained by upsampling the original images. As shown in Table XI, the proposed method can be directly applied to larger inputs without any architectural modification.

Although the computational cost increases with input resolution, the model maintains stable performance across different scales, with only minor fluctuations in F1 scores. Meanwhile, the inference time, throughput, and GPU memory usage scale in a predictable manner and remain within a practically acceptable range. It is worth noting that the cross-correlation operations are performed on intermediate feature maps rather than full-resolution inputs, which effectively limits the additional memory overhead and contributes to the observed efficient scaling behavior. These results demonstrate that PD^{2}Net is not restricted to low-resolution inputs and can be applied to more realistic forensic scenarios involving higher-resolution images.

To further validate the generalization capability of the learned prototypes, we conduct cross-dataset evaluation by training the model on USC-ISI and directly testing it on unseen datasets, including CASIA and CoMoFoD, as shown in Table XII. This setting introduces a clear distribution shift in image content, manipulation patterns, and post-processing conditions.

TABLE X
COMPARISON OF MODEL COMPLEXITY, EFFICIENCY, AND AVERAGE F1 SCORE(%) ON USC-ISI, CASIA, AND COMOFOD DATASETS

Method	Param	Training time	TPS	FLOPS	USC-ISI	CASIA	CoMoFoD
CNN-T GAN	208.32 M	101Mins	2.04	64.78	85.41	27.52	25.00
LBRT	87.76 M	46 Mins	6.21	25.14	79.58	28.12	30.21
CMFDTR	187.64 M	94 Mins	2.93	61.43	80.12	17.49	22.53
PD^{2}Net	158.37 M	28 Mins	6.25	42.96	86.60	30.93	31.86

TABLE XI
HIGH-RESOLUTION INFERENCE ANALYSIS OF PD^{2}NET UNDER DIFFERENT INPUT SIZES

Resolution	Dataset	F1(%)	Time	TPS	GPU (GB)
256×256	CASIA	30.93	0.16	6.25	3.26
512×512	CASIA	29.98	0.58	1.72	6.94
1024×1024	CASIA	28.41	2.21	0.45	13.82

TABLE XII
CROSS-DATASET EVALUATION (F1 SCORE %) OF PROTOTYPE GENERALIZATION UNDER DIFFERENT PROTOTYPE SETTINGS. THE MODEL IS TRAINED ON USC-ISI AND DIRECTLY TESTED ON UNSEEN DATASETS.

Training Set	Test Set	Prototype	Source	Tampered
USC-ISI	CASIA	Ours	30.98	30.87
		Random	27.42	26.95
		Zero	25.86	25.11
USC-ISI	CoMoFoD	Ours	31.64	32.08
		Random	27.91	28.36
		Zero	26.18	26.72

To isolate the contribution of the learned prototypes, we further replace them during inference with randomly initialized prototypes or zero prototypes, while keeping all other network parameters unchanged. As shown in Table XII, both degraded variants exhibit clear performance drops on the two unseen datasets, whereas the learned prototypes maintain stable performance.

These results indicate that the learned prototypes are not merely fitting dataset-specific patterns, but instead provide transferable priors for distinguishing source and tampered regions under different data distributions. In contrast, random or zero prototypes fail to provide such transferable guidance. This provides direct empirical evidence for the prototype generalization capability of the proposed framework.

M. Discussion

Current researchers have proposed universal tampering detection methods that can simultaneously detect manipulations such as splicing, deletion, and copy-move. These approaches primarily rely on detecting anomalous regions based on noise inconsistency, color artifacts, or boundary abnormalities, and are thus effective for identifying externally introduced content. However, in pure copy-move forgery, the tampered region is copied from the same image, introducing only limited inconsistency cues and thereby increasing the difficulty of detection. More importantly, the source region remains an unaltered part of the image and typically does not exhibit discriminative anomaly cues, making it difficult to identify using general-purpose methods. In contrast, PD^{2}Net is designed to explicitly detect and distinguish both source and tampered regions, addressing the core challenge of copy-move localization rather than merely detecting anomalies. Therefore, our method is more specifically tailored to a task that general-purpose methods cannot handle, namely the simultaneous identification and distinction of source-tampered pairs under identical data distributions.

V. CONCLUSION

This paper proposes a generative PD^{2}Net, which differs from the previous discriminative method for copy-move forgery image detection and distinguishment. PD^{2}Net introduces distinct prototypes with the same data distribution to represent the source and tampered region, respectively, solving the challenge of the lack of discrimination clues between the source and the tampered region with high consistency. The prototype is gradually updated and approximated to the target object by step iteration to realize the detection and distinguishment. In addition, to further refine both detection and distinguishment, we introduce the SS and SR modules, which re-detect suspicious regions within the prototype-guided similarity features and effectively suppress false positives. Numerous experiments on USC-ISI, CASIA v2.0, and CoMoFod datasets demonstrate the superiority and robustness of PD^{2}Net. While PD^{2}Net achieves promising results, its increased model complexity introduces additional computational cost. In future work, we plan to explore prototype distillation to transfer the discriminative representation of tampered and source regions from a large teacher model to a compact student network to further reduce model complexity while maintaining accuracy, thereby facilitating real-time and resource-efficient deployment of PD^{2}Net.

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